

A Supplementary Material

A.1 Additional Evaluation Metrics Details

We provide the detailed metric calculation rules in this section. For the Ego4D v1 and Ego4D v2 datasets, our model observes $K = 8$ video segments and generates 5 candidate action sequences with a fixed length $Z = 20$. We compute the Damerau-Levenshtein edit distance (ED) between each candidate sequence and its ground-truth sequence (for verbs, nouns, and actions), reporting the minimum value among the 5 edit distances. For the EGTEA Gaze+ dataset, the model observes the first 25%, 50%, and 75% of the video, predicting the set of verbs that will appear in the remaining portion. The mean average precision (mAP) is calculated for multi-label classification of verbs in three action categories (All, Freq (high-frequency actions), and Rare (low-frequency actions)) to quantitatively evaluate performance.

A.2 Additional Implementation Details

The raw video is sampled at 10 frames per second (fps) before the video description. We employ the VideoLLaMA3-2B model [33] for video description, utilize the Qwen3-max model [31] for real intent extraction, and adopt the Llama3.1-8B-Instruct model [6] for fine-tuning as the action anticipation model. The top_p is 0.8 and the temperature is 0.7 for Qwen3-max, with the future actions as input. LoRA is implemented across all fundamental linear projections within the transformer architecture. In the LoRA configuration, rank (r) is all set to 8, and α is all set to 32. Training is conducted on 2 NVIDIA A100 GPUs.

A.3 Qualitative Analysis of Video Caption Ablation

This section shows a qualitative analysis of LLM performance in action anticipation, comparing video descriptions containing only object and action against those incorporating scene, object, and action, as illustrated in Fig. 5. Adopting 3-dimensional description can perceive environmental characteristics and task-oriented logic, and generate anticipations that fit actual real-world scenarios. In this way, the model can output semantically consistent and task-matched behaviors (e.g., “squeeze cloth” for laundry, “paint wood” for painting), instead of generating generic and repeated action sequences.

A.4 Qualitative Analysis of Intent-Aware Mechanism

In this section, we shows more cases about intent-aware mechanism analysis. The proper intention guidance enables the framework to anticipate more accurate and correct future actions instead of general ones. For instance, intent-aware LLM predicts “pen” instead of “wood” in case 1 (Fig. 6), “flower” rather than “plant” in case 2 (Fig. 7), and “pants” instead of “cloth” in case 3 (Fig. 8).

[illegible]

Fig. 5: Qualitative analysis of video caption ablation. The Scene, Object, and Action description is displayed the key details.

Action sequences of Observed video mark ladder, wipe tape measure, measure ladder, mark ladder, measure ladder, measure ladder, mark ladder, put tape measure
Without Intent-aware mechanism “Predict Actions”: measure ladder, measure ladder, put tape measure, put tape measure, take wood, attach wood, take wood, move wood, move wood, take wood, put wood, adjust wood, take drill, drill screw, put drill, take wood, attach wood, drill screw, put drill, take wood
With Intent-aware mechanism “Predict Intent”: Sequence of actions involving measuring, marking, and assembling with tools and materials “Predict Actions”: put ruler, put pen, take rod, take metal, put metal, take rod, put rod, take metal, put rod, take rod, put rod, take tape measure, pull tape measure, measure rod, measure rod, put tape measure, take rod, move rod, put rod, take rod
GT action sequences put pen, take rod, take rod, put rod, hit rod, put rod, take welding torch, weld ladder, weld rod, put welding torch, take rod, hit rod, put rod, take welding torch, attach rod, take rod, hit rod, hit rod, put rod, take welding torch

Fig. 6: Intent-aware mechanism analysis (Case 1). Blue represents the inferred intent, green indicates the correctly anticipated verbs/nouns, and red is the main discrepancies in noun anticipation with and without intent-aware.

Action sequences of Observed video
cut strawberry, put strawberry, cut flower, put strawberry, take phone, press phone, put phone, detach flower
Without Intent-aware mechanism
“Predict Actions”: take plant , put plant, take plant, put plant, take plant, put plant, take plant, put plant, take plant, put plant, take plant, put plant, take plant, put plant, take plant, put plant, take plant, put plant, take plant, put plant
With Intent-aware mechanism
“Predict Intent”: Planting and arranging flowers with a bag
“Predict Actions”: put flower , take flower , put flower , take flower, put flower , take flower , put flower, take flower , put flower , take flower , put flower , take flower, put flower , take flower, put flower, take flower , put flower, take flower , put flower, put bag
GT action sequences
put flower, detach flower, put flower, put strawberry, detach flower, put flower, detach strawberry, put flower, take flower, put flower, detach flower, put strawberry, detach flower, put strawberry, remove strawberry, put flower, remove strawberry, put flower, remove strawberry, put strawberry

Fig. 7: Intent-aware mechanism analysis (Case 2). Blue represents the inferred intent, green indicates the correctly anticipated verbs/nouns, and red is the main discrepancies in noun anticipation with and without intent-aware.

Action sequences of Observed video
put pants, adjust pants, take iron, iron pants, put iron, adjust pants, take iron, iron pants
Without Intent-aware mechanism
“Predict Actions”: adjust cloth , take iron , iron cloth, put iron, adjust cloth, take iron, iron cloth, put iron, adjust cloth, take iron, iron cloth, put iron, adjust cloth, take iron, iron cloth , put iron
With Intent-aware mechanism
“Predict Intent”: Ironing and adjusting pants multiple times, then handling the iron and cloth
“Predict Actions”: iron pants , put iron , adjust pants , take iron, iron pants , put iron, adjust pants, take iron, iron pants, put iron, adjust pants , take iron, iron pants , put iron, adjust pants , take iron, iron pants, put iron, adjust pants, take iron
GT action sequences
remove pants, put iron, lift pants, adjust pants, put pants, adjust pants, take iron, iron pants, put iron, adjust pants, lift pants, fold pants, put pants, fold pants, lift pants, put pants, take sock, put sock, adjust cloth, take cloth

Fig. 8: Intent-aware mechanism analysis (Case 3). Blue represents the inferred intent, green indicates the correctly anticipated verbs/nouns, and red is the main discrepancies in noun anticipation with and without intent-aware.

A.5 Diversity Analysis

In this section, we present a diversity analysis of the five generated candidate sequences. As shown in Fig. 9, the proposed framework is capable not only of generating different future behaviors based on similar trajectories (e.g., “crafting with clay and sand” vs. “crafting with clay and sand using molds and tools”), but also of deducing multiple distinct future behaviors (e.g., “prepare and clean up after a meal” vs. “prepare and serve coffee with milk”).

GT action sequences of Observed video press clay, wipe floor, move mold, put mold, scrape clay, throw clay, pack sand, move mold Predict Actions Candidate 1: Constructing and shaping materials with molds and tools (Intent) put mold, put sand, turn mold, put mold, cut cement, mix cement, arrange mold, put cement, remove cement, arrange mold, put cement, wipe sand, carry mold, put mold, turn mold, put mold, put mold, cut cement, mix cement, arrange mold, put cement Candidate 2: Crafting with clay and sand (Intent) put clay, put clay, put mold, turn mold, remove mold, pour sand, pour sand, put mold, arrange sand, cut, clay, roll clay, pull mold, throw, clay, adjust clay, scrape clay, throw clay, apply sand, lift mold, put mold, turn put mold Candidate 3: Constructing and shaping materials with molds and tools (Intent) move mold, put sand, turn mold, hit mold, turn mold, cut cement, roll cement, take cement, move mold, put cement, mix mold, cut cement, take cement, throw cement, clean sand, move mold, turn mold, put mold, turn mold, move mold Candidate 4: Crafting with clay and sand using molds and tools (Intent) put clay, press clay, adjust clay, put clay, move mold, turn mold, remove mold, pour sand, pour sand, hit mold, put mold, clean floor, cut clay, roll clay, roll clay, move mold, put clay, press clay, adjust clay, throw clay Candidate 5: Constructing and handling molds with clay and sand (Intent) move mold, turn mold, put mold, pack sand, pour sand, put mold, pack clay, press clay, put clay, press clay, pack sand, move mold, turn mold, put mold, pack sand, pour sand, put mold, pack clay, press clay, put clay GT action sequences turn mold, hit mold, pack sand, put mold, clean sand, pack soil, adjust mold, put clay, scrape clay, press clay, adjust mold, move mold, turn mold, clean sand, pack sand, pour sand, clean sand, pack clay, press clay, adjust mold	GT action sequences of Observed video wash bottle, lift bottle, take sponge, wash bottle, close faucet, lift bottle, put bottle, take sieve Predict Actions Candidate 1: Prepare and clean up after a meal (Intent) put pot, open cabinet, take cup, close cabinet, open bag, put bag, take lid, take lid, take lid, take lid, put lid, take lid, put lid, take lid, put lid, take lid, take lid Candidate 2: Prepare and serve coffee with milk (Intent) put pot, close cabinet, open cabinet, take cup, open cabinet, put cup, close cabinet, cabinet, take jug, open fridge, take milk, open milk, pour milk, put container, put container, close fridge, open cabinet, take cup, put cup, take cup Candidate 3: Prepare and serve coffee (Intent) take bag, put bag, open bag, pour coffee, put bag, take jug, open faucet, wash jug, close faucet, pour water, put jug, take lid, take lid, cover lid, take jug, open faucet, close faucet, pour water, put jug, take lid Candidate 4: Prepare and serve a beverage (Intent) take cup, open cabinet, put cup, close cabinet, open cabinet, take cup, jug, pour beverage, put cup, take cup, put cup, take cup, put cup, take cup, put cup, take cup, put cup, take cup, put cup, take cup, put cup Candidate 5: Prepare and clean up after a meal (Intent) put pot, take lid, put lid, take paper, open paper, take container, put paper, take container, open container, put lid, take cup, take cup, open kettle, put container, open kettle, put container, open cabinet, take plate, put plate, put plate GT action sequences adjust stove, squeeze sieve, touch faucet, open faucet, close faucet, wipe towel, take timer, put watch, take milk, clean floor, put take, wash lid, close faucet, put sponge, shake lid, take wine, take counter top, hold tea, open report
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Fig. 9: Diversity analysis of 5 generated candidate sequences. Blue represents the inferred intent, and green indicates the correctly anticipated verbs/nouns.

A.6 Additional Discussion

Recent LTA methods have witnessed significant progress of integrating LLMs and intention reasoning. Methods like INSIGHT [4] have demonstrated the value of explicit cognitive reasoning through hand-object features and reinforcement learning-based intention inference. However, these approaches share a fundamental limitation: cascaded multi-stage pipelines that separate action recognition from temporal reasoning. This design inherently suffers from error propagation—recognition errors from the first stage corrupt subsequent reasoning, particularly for rare actions and out-of-distribution scenarios.

Moreover, the use of online reinforcement learning (e.g., INSIGHT [4]) for intention inference introduces additional challenges. The training process requires careful hyperparameter tuning which raise concerns about sample efficiency and training stability. In addition, the reinforcement learning objective may encourage the model to exploit spurious correlations in the training data, leading to overfitting problem. The separation of modules also precludes joint optimization of visual representation learning and temporal reasoning, leaving potential performance gains unrealized.

In contrast, the proposed Intent-Aware Structured (IAS) framework adopts a unified end-to-end architecture that eliminates error propagation entirely. By

fine-tuning a single LLM with parameter-efficient LoRA, we achieve stable training without RL volatility. Using pre-extracted real intents as supervision avoids the sample inefficiency and overfitting risks inherent to RL-based intention learning. This holistic optimization yields robust generalization and competitive performance.

In summary, IAS offers a more robust and unified alternative to cascaded RL-based approaches. Nevertheless, several avenues for future work remain. First, integrating the process of structured video captioning into an end-to-end trainable framework could further enhance performance. Second, inferring high-quality intentions directly from observed video—without access to future ground truth could be an important direction for making intent-aware anticipation more practical. Finally, extending the IAS framework to handle even longer prediction horizons and complex hierarchical action structures could unlock new applications in more areas.